

# Sienna Camper Conversion

A modular, tool-free lift-out sleeping + galley system for a Toyota Sienna — sleeps two, with a slide-out fridge and camp kitchen, EcoFlow power and A/C, and no permanent modification to the van. This page summarizes every feature and shows the drawings, so you can skim it and flag anything you'd change.

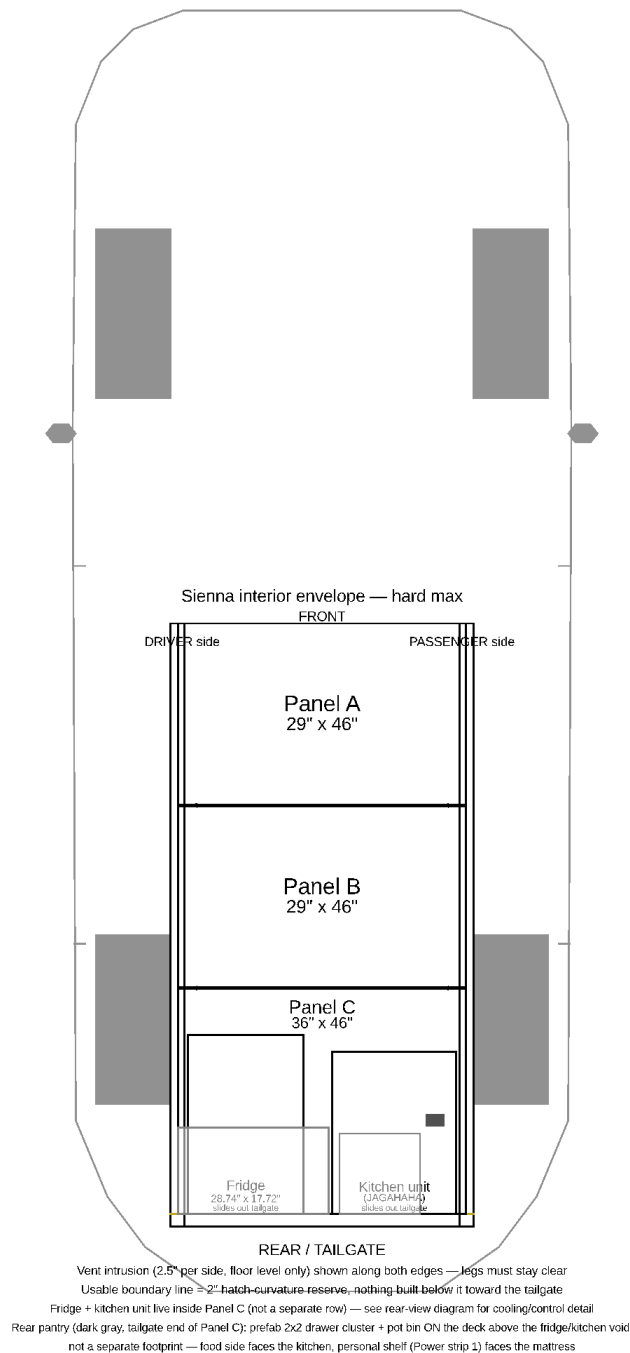
**How to give feedback:** skim the seven drawings and the feature list, then jump to [Open questions](#) — those are the calls I'd most like a second opinion on (and the measurements still unverified). Tell me what's missing, what you'd do differently, or where the design looks fragile.

|                                                              |                                                                                      |                                                                       |                                                                |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| <b>SLEEPS</b><br><b>2</b><br>52"-wide platform, 25"/person   | <b>VAN INTERIOR</b><br><b>96 × 48.5 × 44"</b><br>L × W × H (length est.)             | <b>MODULES</b><br><b>3 + 4</b><br>Panels A/B/C + prefab drawer pantry | <b>POWER</b><br><b>DELTA 3 +</b><br>Extra Battery · WAVE 3 A/C |
| <b>COLD</b><br><b>41 QT</b><br>Rocky 40 dual-zone, slide-out | <b>EST. COST</b><br><b>\$3.4–3.7k</b><br>full incl. spare kit; ~\$2.7k if gear owned |                                                                       |                                                                |

## THE LAYOUT

One continuous deck of three lift-out boxes; the fridge, kitchen, and pantry all share Panel C at the tailgate end.

## BIRD'S-EYE FLOOR PLAN

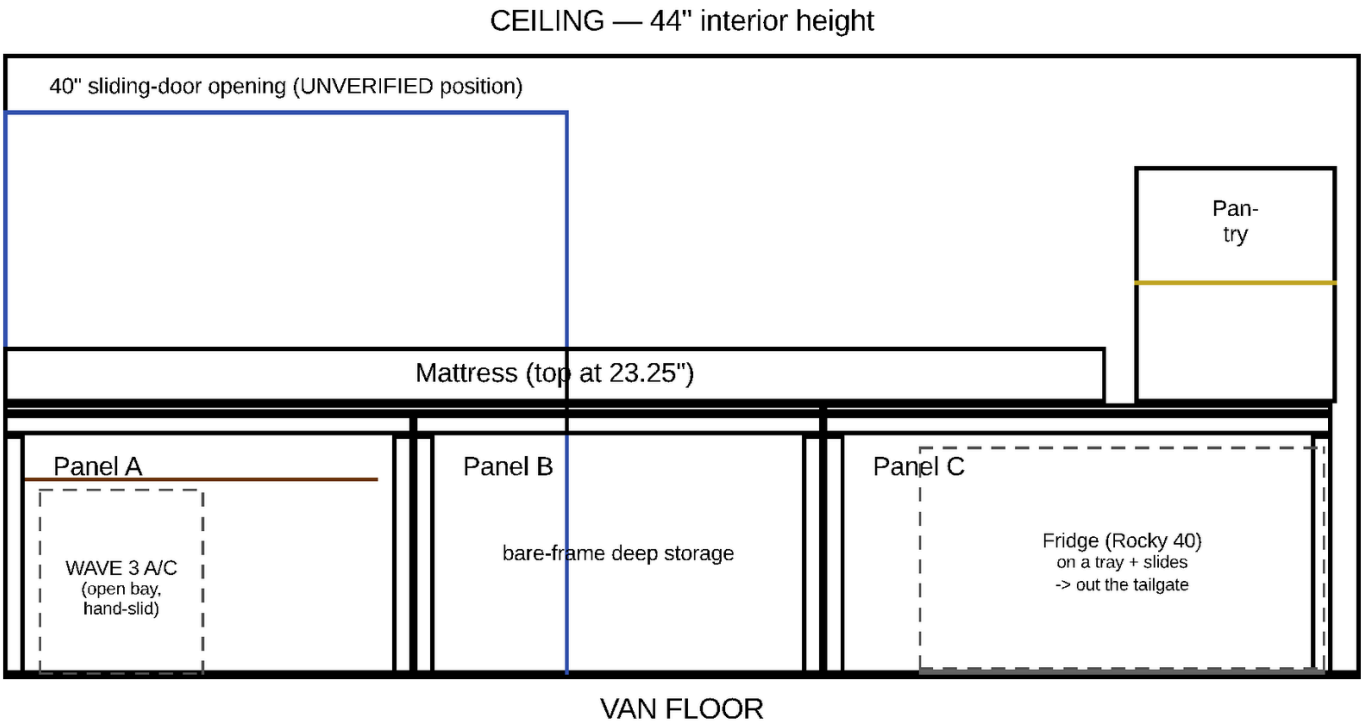


**Top-down.** Front of the van at top, tailgate at bottom. Panel A (29") + Panel B (29") carry the bed; Panel C (36") holds the fridge (driver side) and kitchen (passenger side) underneath, with the prefab drawer pantry on the deck above them. Grey blocks = wheel wells; the platform stays clear of the floor-level heat vents at the edges.

## THE DRAWINGS

The two side elevations show what each sliding door reaches; the two gate views show the galley stowed for driving vs. deployed for use.

DRIVER SIDE

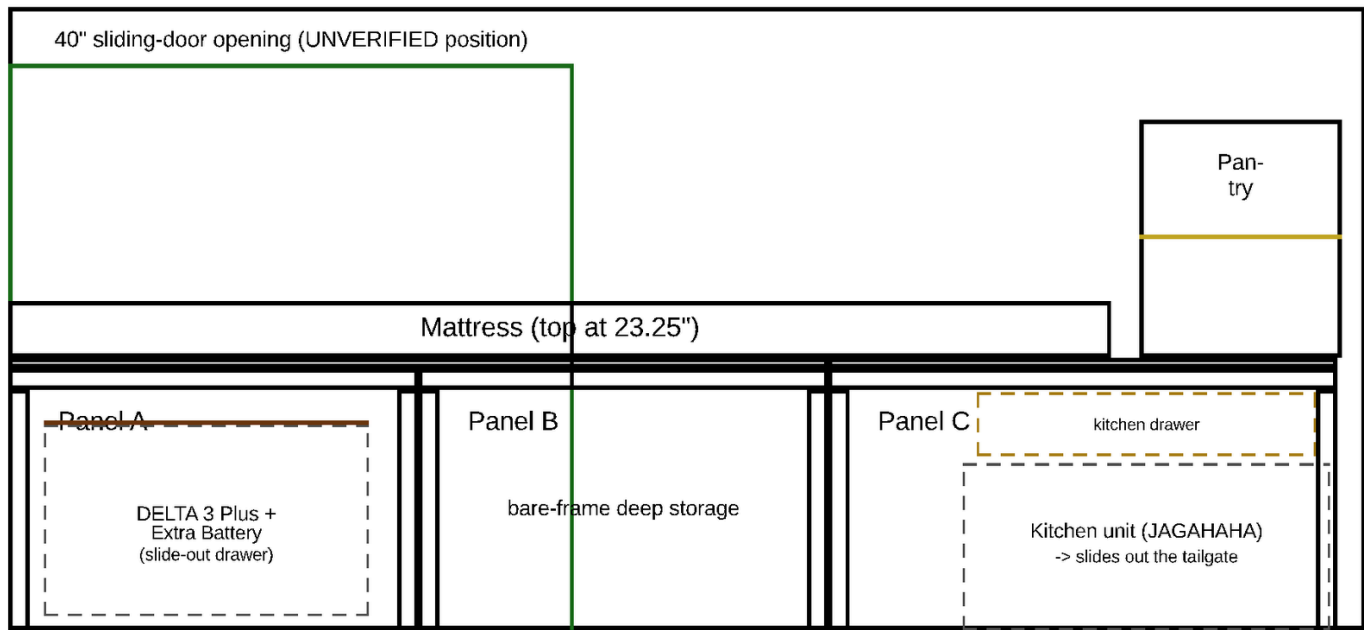


DRIVER SIDE — looking in from this door. FRONT SEATS at left, TAILGATE at right.  
This side: WAVE 3 (Panel A) + fridge (Panel C). Dashed = under-deck items reached from here.

**From the driver door.** WAVE 3 A/C in Panel A's open bay; the fridge under Panel C. Dashed = under-deck items reached from this side.

## PASSENGER SIDE

CEILING — 44" interior height



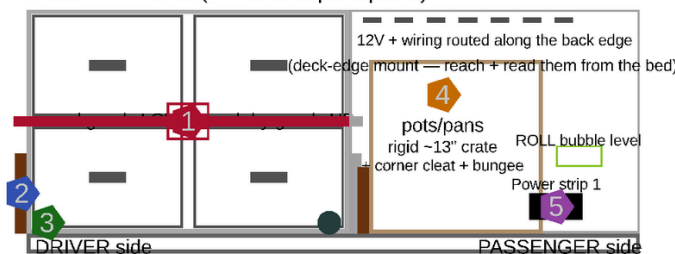
### VAN FLOOR

PASSENGER SIDE — looking in from this door. FRONT SEATS at left, TAILGATE at right.  
This side: DELTA 3 power (Panel A) + kitchen unit (Panel C). Dashed = under-deck items reached from here.

**From the passenger door.** The DELTA 3 power drawer in Panel A; the kitchen unit + its hung drawer under Panel C.

## REAR PANTRY — PREFAB DRAWER CLUSTER

2x2 IRIS 12"W stackable drawers ~21.8" open deck bay  
24.2" W x 16.8" H — 4 units (2x Home Depot 3-packs)



### Hold-down + fit-out

- 1 Cam strap across drawer fronts (shut + hold-down)
- 2 Birch cleats: cab side + both sides (tailgate open)
- 3 Flush deck D-rings (strap anchors)
- 4 Pot/pan crate (~13") in the open bay
- 5 Power strip 1 + ROLL level, relocated to the deck

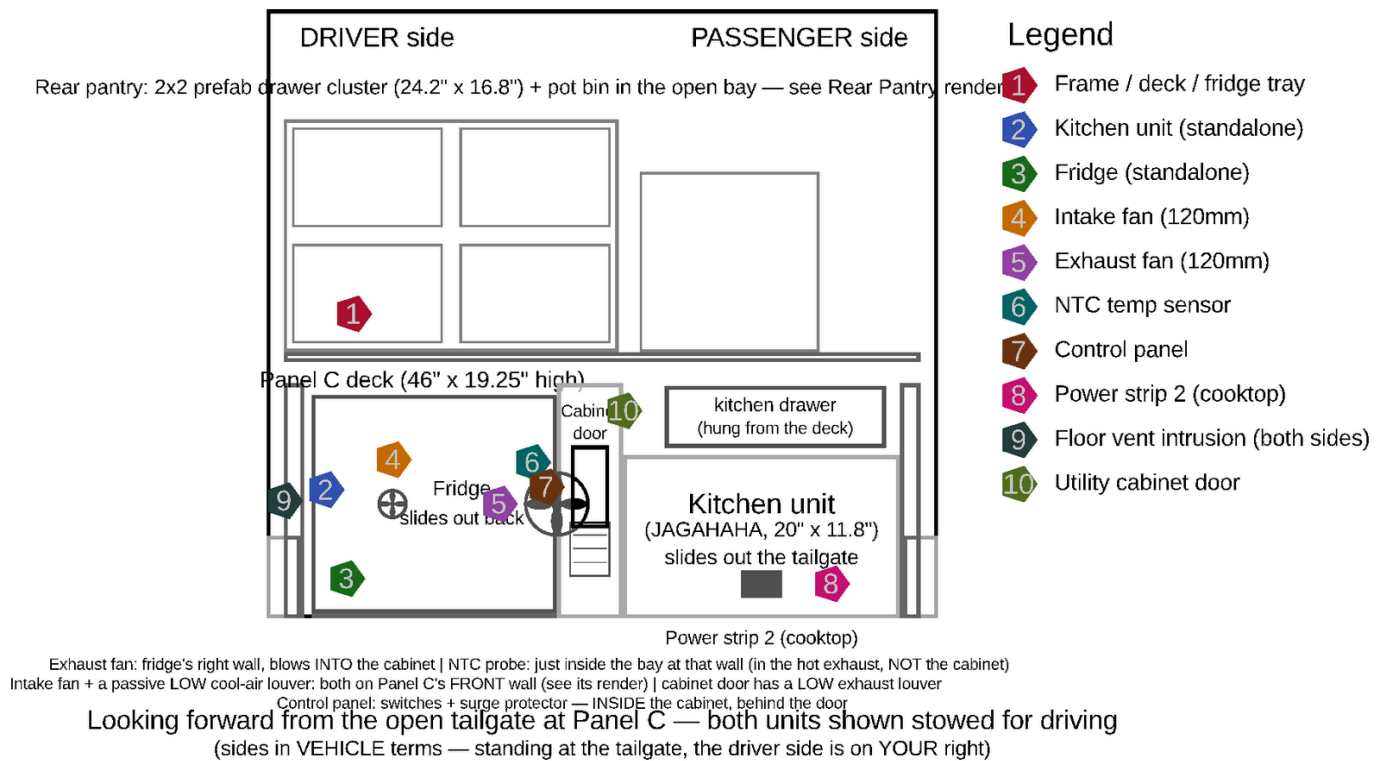
Panel C deck (46" wide, tailgate end) — cluster + bay just SIT on it, held by cleats + a strap

**REAR PANTRY** — prefab 2x2 drawer cluster + pot bay (replaces the plywood pantry; ~29 lb lighter, nothing built)  
Each drawer unit lifts straight out (loosen the strap) — removable, clears the liftgate. Cab-side cleat stops the forward slide under braking.

**Bought, not built.** The pantry is a 2x2 stack of IRIS 12"W drawers (24.2" x 14.3" x 16.8", ~15 lb) on Panel C's deck, plus a ~13" pot crate in the leftover 21.8" bay. A cleat pocket + one cam strap hold everything; each drawer unit lifts straight out. Power strip 1 + the roll bubble level mount on the deck edge beside it.

## INTO THE GATE — STOWED

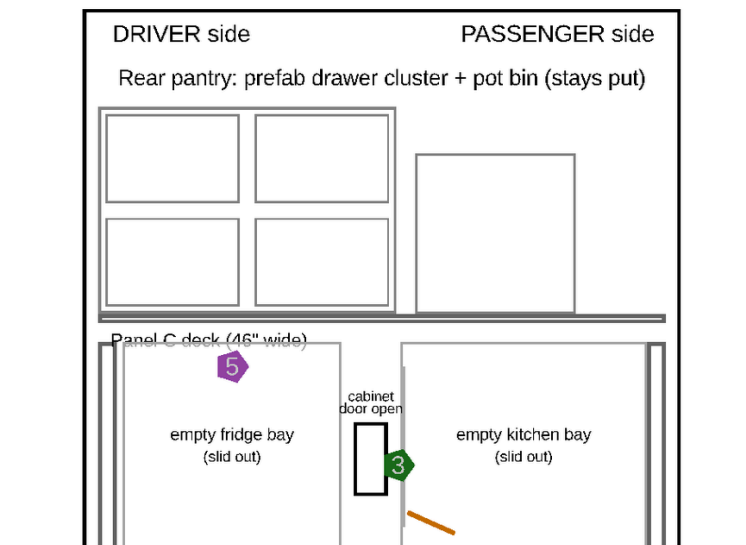
Sienna interior envelope (width x height) — hard max



**Driving mode.** Fridge + kitchen tucked under Panel C's deck; cooling fans, NTC sensor and control panel live in the utility cabinet between them.

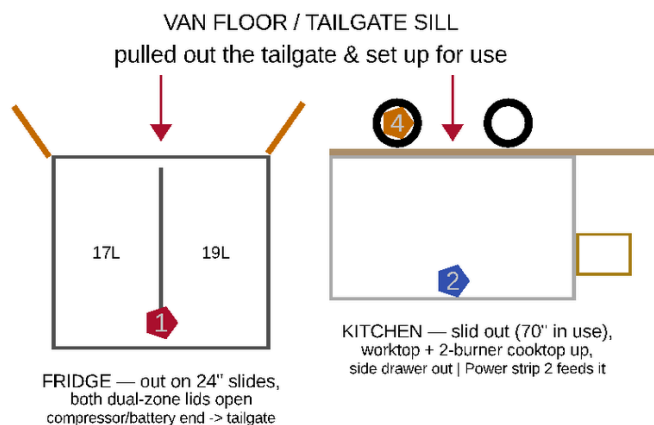
## INTO THE GATE – DEPLOYED

Sienna interior envelope (width x height)



### Legend

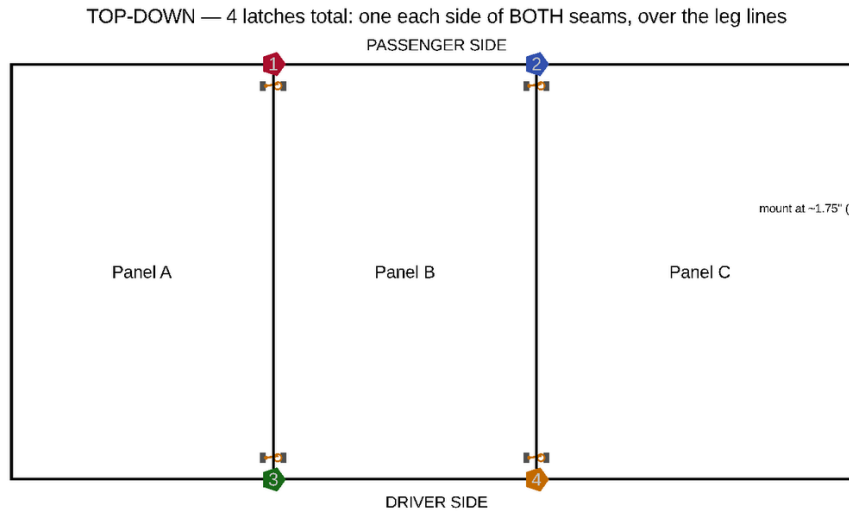
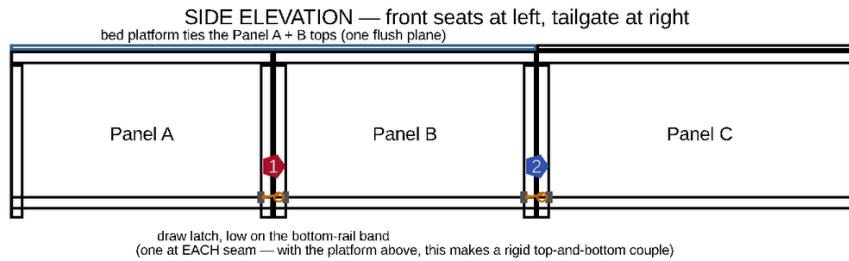
- 1 Fridge — deployed, both lids open
- 2 Kitchen unit — deployed, cooktop up, drawer out
- 3 Utility cabinet (door open): control panel + fans
- 4 2-burner induction cooktop on the worktop
- 5 Empty bays left behind in Panel C



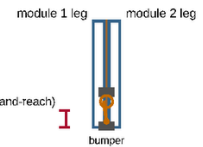
DEPLOYED — looking into the open tailgate with the fridge & kitchen pulled out and set up for use

**Camp mode.** Both slide out the tailgate: fridge lids open (dual-zone 17L+19L), kitchen worktop + 2-burner cooktop up and drawer out. The bays behind them sit empty.

## SEAM DRAW-LATCH POSITIONING



DETAIL — over-center draw latch  
base plate on one leg, hooked bail + lever on the other;  
flip the handle to clamp tight, flip to release (no tools)



SEAM DRAW-LATCHES — clamp the 3 lift-out modules into one rigid beam (4 latches: both sides of the A/B and B/C seams)  
Alignment pins LOCATE the modules; the draw latches PULL them tight against the bumper. Hand-released — modules still lift out in seconds.

**Clamping the modules.** The alignment pins only *locate* the three boxes; four hand-released over-center draw-latches — one on each side of both seams, low on the bottom-rail band — pull them tight into one rigid beam. With the bed platform tying the tops, the low latches complete a top-and-bottom couple that kills sway and rattle, while any panel still lifts out in seconds.

## FEATURES

What each part is and why it's there. Amber notes flag the ones I'd most like feedback on.

## STRUCTURE **Modular lift-out platform**

Three independent boxes — Panel A (29"), B (29"), C (36"), all 46" wide — built on 2×2 pine frames with leveling legs, plus a clamp-on pantry. Nothing bolts to the van; each box lifts out through the liftgate as a one/two-person carry. Cube-frame bottom rails add racking stiffness, and **4 hand-released over-center draw-latches** at the seams clamp the three boxes into one rigid beam without giving up the lift-out.

Built weight is  $\approx$  **441 lb** (incl. an interior spare tire at the axle) (after lighter-wood swaps), ~884 lb loaded with two people — comfortably under the van's payload, though the mass is rear-biased (see the weight budget in the plan).

## SLEEPING **Cantilevered bed platform**

A 52"-wide slatted 1×4 platform spans Panels A+B and ends flush at the B/C seam, meeting Panel C's own deck. It overhangs the 46" boxes by 3" per side — using the fact that the van is wider up at mattress height than at the pinched floor.

**Feedback wanted** — this depends on  $\geq 53"$  of wall-to-wall width at 19–23" above the floor, which isn't measured yet (see question 3).

## SLEEPING **Mattress**

HEST Dually Long — 78 × 50 × 4" solid foam with a washable waterproof cover, 25" per person. Leaves roughly 16.75" of sitting headroom to the ceiling.

## GALLEY **Rear pantry — prefab drawers**

Storage is bought, not built: four IRIS USA 12"W stackable drawers (Home Depot 3-packs) stacked 2×2 on Panel C's deck at the tailgate — 24.2" wide, 16.8" tall, ~15 lb — plus a rigid pot crate in the remaining ~22" of deck, and 4 Sterilite 28-Qt lidded totes containerizing Panel B's top-load bay. A screwed-down cleat pocket stops it sliding under braking, and one cam-buckle strap across the drawer fronts keeps the drawers shut and the stack snug; loosen it and any unit lifts out (~3 lb each). No joinery, no retention hardware, and the drawers contain their own contents.

**Feedback wanted** — would you trust fabric/plastic prefab drawers + a strap on washboard roads, or is the strapped-down rigidity worth checking on a rough test drive first?



### **GALLEY** Fridge

BougeRV Rocky 40 — dual-zone 41 QT, 40.6 lb, 60W max — on a heavy-duty 24" slide + plywood tray, pulled out the open tailgate for lid access. The compressor/battery end faces the tailgate so the optional battery swaps without a full slide-out.

### **GALLEY** Kitchen unit

JAGAHAHA slide-out camp kitchen (2-burner, own drawer), strapped to floor anchors, sliding out the tailgate and extending to ~70" in use. A shallow kitchen drawer hangs in the dead air above it for utensils and flat goods. The induction cooktop runs off Power strip 2.

### **COOLING** Forced-air fridge bay

Two 120mm fans — intake on Panel C's front wall, exhaust on the fridge's kitchen-facing wall blowing into the utility cabinet — on a W1209 thermostat reading an NTC probe in the hot exhaust, so they only spin when the compressor is working. Two passive louvers (low intake + cabinet exhaust) back them up. This compensates for the bay being tighter than the fridge's passive-venting spec.

### **POWER** EcoFlow system

DELTA 3 Plus + Smart Extra Battery ride in Panel A's slide-out drawer (charged at camp via shore power/solar, not from the van). The WAVE 3 air conditioner stores in Panel A's open bay and runs at the tailgate or the front seat. Two AC power strips (cooktop + pantry) plug into the van's REAR outlet; the front console outlet (1500W verified) charges the DELTA 3 — all sharing one 1500W inverter budget.

### **LEVELING** Two-stage leveling

Level the van at the wheels with Lynx blocks, guided by a phone Block Calculator that reads two bed-mounted bubble levels and tells you blocks-per-tire. The interior leg feet are then set once against the van's own floor. (Electric feet were considered and rejected.)

**STORAGE****Found space + joinery**

Reclaimed dead headroom — a tray over the DELTA 3 stack, a shelf over the WAVE 3, bins in the cabinet gap — and Panel B is a bare-frame deep-storage cube. The bottom stabilization frame was dropped lower for more usable volume. Carcasses use Ryobi DBJ50 biscuit joints (R-series) for strength and clean faces, with fastener locations dimensioned.

**SAFETY****Air & alarms**

An owner-placed low-mounted CO monitor (audible from the bed) and a fire extinguisher, plus SRS seat-delete emulators so the airbag system reads normally with the 2nd row removed.

**OPEN QUESTIONS FOR REVIEWERS**

The unverified measurements and judgment calls. Items tagged **VERIFY** need a tape measure on the actual van before anything is cut.

**1 Interior length** **VERIFY**

96" is an estimate for the empty interior with the 2nd row removed. Measure closed-hatch to front seatbacks — the whole assembly is sized to fill it exactly.

**2 Liftgate opening** **VERIFY**

48 × 36" is web-sourced, not measured. Every module must pass through it upright, and the upper corners round off — measure the real narrowest width and height.

**3 Bed cantilever width** **VERIFY**

The 52" platform overhangs the 46" boxes and needs ≥53" wall-to-wall at 19–23" above the floor along the whole run. If it's tighter, the platform shrinks.

**4 Side-door position & reach** **VERIFY**

The door's fore-aft position is a placeholder. It decides which drawers are actually reachable from the side vs. blocked by body structure.

#### 5 **Rear-corner floor vents** VERIFY

Panel C's rear legs sit at the true corners; confirm the floor heat vents don't reach the last ~4" there.

#### 6 **Pantry cluster vs. liftgate swing** VERIFY

The 18.6" drawer cluster + pot bin sit at the tailgate. Confirm the gate's swing-up clears them and the fridge/kitchen still slide out past.

#### 7 **Appliance specs & price** VERIFY

Confirm live-listing dimensions and current price for the Rocky 40 fridge and the JAGAHABA kitchen before ordering.

#### 8 **Floor anchors**

The E-track anchors bolt through the floor pan — inspect underneath for fuel/brake lines and wiring, and seal each hole.

### DESIGN CALLS WORTH A SECOND OPINION

Beyond the measurements, these are open on the merits — I'd value a view:

- Is a **slide-out kitchen + separate fridge** the right split, or would one integrated galley be simpler and stronger?
- Is **two 120mm fans + passive louvers** enough fridge cooling for real summer heat, or should the bay get a bigger fan or a dedicated vent?
- Prefab drawer pantry at the tailgate vs. moving it forward — does having it right at the gate get in the way in practice?
- Anything **missing entirely** — water/sink, interior lighting, a shore-power inlet, insulation, screens?

### COMPANION TOOLS

The interactive/printable sheets that go with this plan.

### **Block Calculator** →

Blocks-per-tire from the bubble levels

### **Shopping List** →

Materials, quantities & links

### **Cut List** →

Every board & panel, grouped

### **Build Sequence** →

Dependency-ordered assembly

Drawings are to-scale OpenSCAD line art from the parametric model; every dimension is computed from the plan's source of truth, not sketched. Full detail lives in **Project\_Smores.pdf** (Components 1–10 + appendices). Estimates and unverified figures are flagged throughout.